

ASHISH JOSHI

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PROFESSIONAL SUMMARY

Data Science and Machine Learning-focused undergraduate with hands-on experience in Python, data analysis, machine learning, and deep learning. Experienced in developing end-to-end projects using Scikit-learn, XGBoost, PyTorch, and TensorFlow, including CNN-based image classification and U-Net-based MRI tumor segmentation. Skilled in data preprocessing, feature engineering, model optimization, and performance evaluation.

TECHNICAL SKILLS

Programming & Databases: Python, SQL (basics — queries, filtering, joins, aggregation)

Data Science & Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Missing-Value & Outlier Handling, Feature Engineering, Feature Selection, Categorical Encoding, Feature Scaling, Imbalanced Data Handling (SMOTE), Time Series Analysis (Statsmodels, Prophet), Web Scraping (BeautifulSoup, Requests)

Machine Learning: Scikit-learn, Regression, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, KNN, SVM, K-Means, Cross-Validation, Hyperparameter Tuning

Model Evaluation: Classification Metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC, PR-AUC), Regression Metrics (RMSE, MAE, R^2), Segmentation Metrics (Dice Score, IoU), Confusion Matrix

Deep Learning & Computer Vision: PyTorch, TensorFlow/Keras, CNN, U-Net, VGG16, MobileNetV2, Transfer Learning, Fine-Tuning, Image Classification, Semantic Segmentation, Image Augmentation, OpenCV, BCE + Dice Loss, CUDA/GPU Training

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau

NLP: NLTK, Text Preprocessing, Tokenization, Stopword Removal, Stemming & Lemmatization, TF-IDF, Bag-of-Words, Text Vectorization

Deployment & Tools: Streamlit, Hugging Face, Git/GitHub, Jupyter Notebook, Google Colab, VS Code, Joblib

PROJECTS

Brain Tumor Segmentation from MRI Images | PyTorch, U-Net, VGG16, Albumentations, OpenCV, NumPy

[github.com/Ashishjoshi794/EDA-PROJECT\(u_net_and_vgg16.ipynb\)](https://github.com/Ashishjoshi794/EDA-PROJECT(u_net_and_vgg16.ipynb))

- Developed a binary semantic segmentation pipeline in PyTorch on the LGG brain MRI dataset, validating 3,929 image-mask pairs (34.95% tumour-positive) and building a custom Dataset class with stratified 70/15/15 splits.
- Preprocessed and augmented TIFF MRI slices with Albumentations (128×128 resizing, flips, ±15° rotation, ImageNet normalisation) and trained using a combined BCE + Dice loss and Adam on GPU.
- Implemented and compared three architectures: a U-Net built from scratch, a VGG16 encoder-decoder, and a hybrid model pairing an ImageNet-pretrained VGG16 encoder with U-Net skip connections.
- Evaluated all models on held-out test data using Dice, IoU, precision and recall; the VGG16 + U-Net hybrid performed best (Dice 0.912, IoU 0.886, Precision 0.943, Recall 0.934), a 13.6% relative Dice improvement over the U-Net baseline.

Cat vs Dog Image Classification | Python, TensorFlow, Keras, MobileNetV2, Streamlit

github.com/Ashishjoshi794/Dog-and-Cat-Classification

- Built a binary image classifier on the Kaggle Cat vs Dog dataset (24,961 images) by fine-tuning a pre-trained MobileNetV2 CNN, unfreezing the final 30 layers and training with RMSprop and dropout regularisation.
- Implemented an input pipeline resizing images to 224×224 RGB with colour standardisation matching MobileNetV2 requirements; achieved 98.90% accuracy, 98.62% precision, 99.20% recall and 98.91% F1-score.
- Deployed the model as a multi-page Streamlit app for real-time image upload and prediction, with confidence scores and a 70% threshold that flags out-of-distribution images instead of forcing a label.

Employee Attrition Prediction | Python, Pandas, Scikit-learn, XGBoost, Streamlit, Joblib

github.com/Ashishjoshi794/Employee-Attrition-Prediction

- Developed an end-to-end employee attrition classification pipeline using 1,470 IBM HR records and 16 selected features.
- Built a Scikit-learn preprocessing pipeline and tuned Logistic Regression, Random Forest, and XGBoost using 5-fold GridSearchCV with F1-score optimization.
- Selected Logistic Regression, achieving **75.51% Accuracy, 63.83% Recall, 45.45% F1-score, and 76.58% ROC-AUC**.
- Deployed the final model with Joblib and Streamlit, providing attrition predictions, stay/leave probabilities, and model comparisons.

EDUCATION

Bachelor of Cyber Security (BCS.IT)

KFA Business School, Kathmandu, Nepal | Currently studying — 8th Semester

+2 (Higher Secondary), Science / Computer Science — Golden Gate International College, Kathmandu, Nepal | Completed

Certifications / Training (*Certificates on Portfolio*)

Data Science Course — Mind risers Institute of Technology

Machine Learning Using Python and **Introduction to Data Science** — Simplilearn SkillUp